Reproduction of Self-Supervised Poisson Denoising: The Poisson2Sparse Framework

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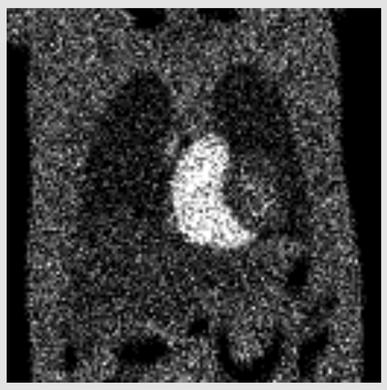
1. Introduction

Problem.
Biomedical images are corrupted by signal-dependent Poisson noise, while clean ground truth is rarely available.
Solution.
We reproduce *Poisson2Sparse*, a self-supervised deep unrolling framework that learns directly from single noisy images.
Contribution.
The full pipeline is ported to Apple Silicon (M4 / MPS) for efficient GPU-accelerated training.

2. Research Objective

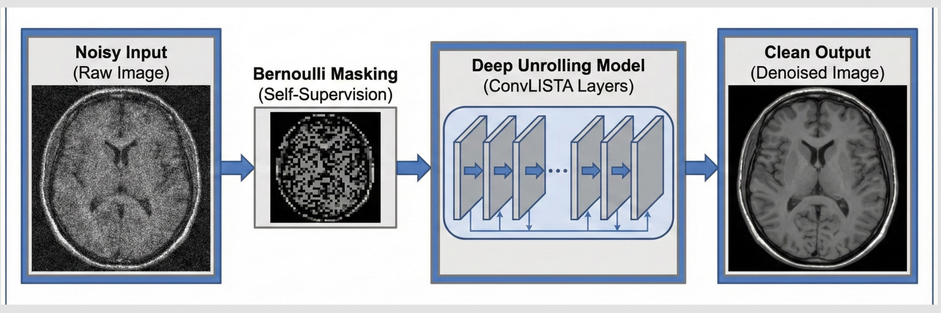
Our study has four goals:
1.Re-implement the ConvLISTA model,
2.Run it on Apple Silicon via the MPS backend,
3.Verify anatomical fidelity using method-noise,
4.Reach ~31.9 dB PSNR, approaching high-end GPU performance on consumer hardware.



3. Related Literature

Poisson2Sparse [Ta et al., 2022]: Self-supervised framework combining sparse representation with deep unrolling for Poisson noise.
BM3D [Dabov et al., 2007]: State-of-the-art non-local collaborative filtering; requires VST for Poisson noise.
Neighbor2Neighbor [Huang et al., 2021]: Self-supervised denoising using sub-sampling from single noisy images.
LISTA [Gregor & LeCun, 2010]: Introduced the concept of unrolling iterative algorithms into neural networks.

4. Design



4.1 Realistic constraints and conditions

Apple M4 uses much **less energy than data-center GPUs**, making the system more environmentally friendly.

4.2 Cost of the design

The system is designed to run efficiently on a consumer-grade processor using open-source software and publicly available datasets, allowing the entire denoising pipeline to be developed and trained without relying on specialized server hardware or proprietary tools.

4.3 Engineering Standards

The code follows PEP 8 for clean and maintainable development and IEEE 29119 for systematic testing, while the overall design aligns with DICOM standards to ensure compatibility with medical imaging systems.

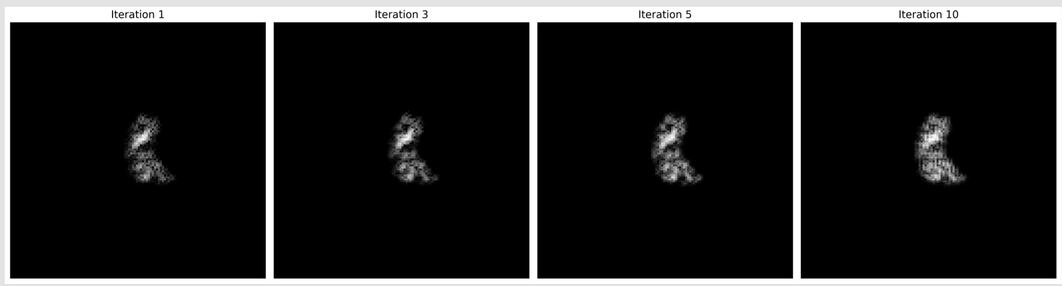
4.4 Details of the design

The system uses deep algorithm unrolling, where the ISTA optimization process is turned into a 10-layer ConvLISTA network that progressively refines the image. An encoder–decoder structure with 512 small convolutional filters learns to separate anatomical details from noise. Training is **self-supervised** by splitting a single noisy image into two complementary views, removing the need for clean reference data. The pipeline is optimized for Apple Silicon using the **MPS backend** and unified memory for efficient computation.

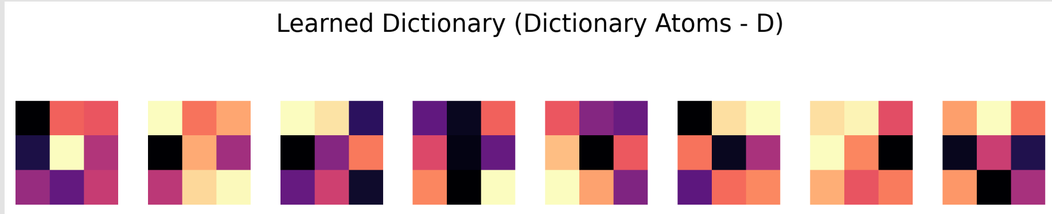
5. Methods

The design turns the classical ISTA solver into a 10-layer ConvLISTA network, where each layer acts like one optimization step:

- **Reconstruct**
- **Compare with noisy input**
- **Update sparse code**



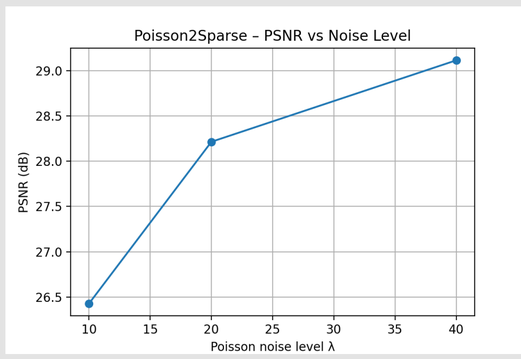
An encoder–decoder with 512 small filters learns anatomical patterns, while training is guided by a Poisson-based loss that models signal-dependent noise.



Self-supervised training uses two complementary views from one noisy image, while sparsity and regularization preserve anatomical structure.

6. Results and Discussions

A) Quantitative Trend
Theoretically, PSNR is expected to increase as λ increases; however, since the results for λ=10 and λ=40 are reported based on a reduced 1,000-iteration training schedule, the direct comparison is limited.

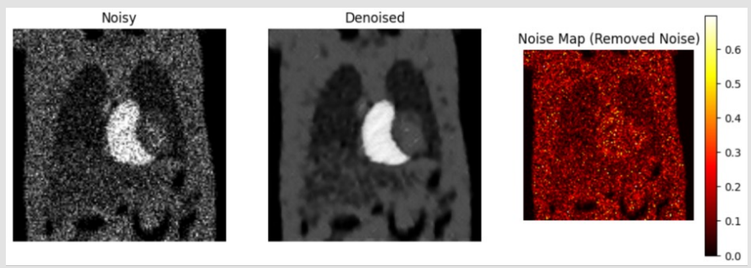


***B)* Comparative Performance**

Noise Level (λ)	Original Paper (GPU)	Our Reproduction (Apple M4)
λ= **40** (Low Noise)	34.31 dB	**29.12 dB**
λ= **20** (Med Noise)	32.20 dB	**31.88 dB**
λ= **10** (High Noise)	30.00 dB	**26.43 dB**

Original values are taken from Poisson2Sparse (Ta et al., 2022); reproduction values are obtained from our Apple M4 implementation.

C) Visual Fidelity
The model successfully removes Poisson noise while preserving key anatomical structures.



7. Conclusion

*By reproducing Poisson2Sparse, we show that self-supervised deep unrolling achieves high-quality denoising (*31.88 dB PSNR*) on portable hardware, preserving anatomical details without the need for server-class GPUs. Future work will focus on clinical use and further optimization.*

Acknowledgements

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